Superdeterminism in a Block Universe: The Case for a Common Generator

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Abstract

The particle source and the devices choosing measurement settings are parts of one world. A short common rule could coordinate them without listing every outcome separately. This paper develops that superdeterminist proposal using a finite deterministic example. To reproduce Bell-inequality violations locally with definite outcomes, such a model must link the measurement settings to the hidden information determining the outcomes. This does not require the two setting devices to be statistically dependent on each other. The example makes the distinction explicit and shows why changing how states are prepared changes the correlations. The target is a common generator that derives the full quantum statistics. Determinism supplies neither those statistics nor the required dependence by itself.

Keywords: superdeterminism, measurement independence, Bell correlations, description length, model comparison, fine tuning

1. Specificity is not complexity

A billion specified digits can come from a short generating rule. The digits of pi illustrate the point: a finite prefix can be described by an algorithm together with the requested precision. Its description need not list every digit. Specificity is not complexity.

This observation answers a specific objection to superdeterminism. Correlations between settings and measured systems need not be separately entered in a cosmic lookup table. A shared generating rule can coordinate them. The task is to find a rule that explains the observed coordination, charging the rule, its boundary and its observation measure together.

Fine tuning has several meanings. It can concern the length of a description, sensitivity to perturbations, small measure of admissible preparations, or a causal model's failure of faithfulness. Refuting an inference from specificity to long description does not answer all four. Wood and Spekkens [1], for example, formulate a problem involving observed independence and causal parameters. Their result is not the claim that each initial particle coordinate must be listed separately.

Matheism puts the source, apparatus, setting devices and observer inside the same mathematical world. This motivates a common generator account in which experiments are physical processes throughout. The generator must explain

both what is measured and how the measurement conditions arise. Its coordination may be simple even when no observer can predict the individual results.

A mathematical derivation establishes its conclusion from its premises without experimental confirmation of that implication. A priori arguments for Matheism must be assessed for validity and the justification of their premises; being philosophical does not make them mere empirical conjectures. Deriving quantum statistics from a specified generator is also a mathematical task. Identifying that generator and its assumptions with our observed world is a further claim unless the argument independently establishes that identification. The distinction is between proof, application and empirical discrimination, not between strong mathematics and weak philosophy.

2. The conditional case for superdeterminism

Let a and b denote two settings, A and B the outcomes, and lambda the model's relevant underlying state. Measurement independence is the condition P(lambda | a,b) = P(lambda). Local factorization is P(A,B | a,b,lambda) = P(A | a,lambda)P(B | b,lambda).

The position defended here takes deterministic local causation and definite outcomes as starting commitments. Given Bell violating correlations, this makes measurement dependence necessary within the stated hidden variable framework. A common past generator is then a natural first candidate: the source and the setting devices inherit their coordination from physical conditions in the same world. This is a conditional preference with a theorem behind the required dependence. The theorem does not select a unique generator or establish that every alternative is logically impossible.

Locality here means the displayed factorization, not merely the absence of controllable faster than light signalling. Likewise, the block view establishes that settings belong within the world; Bell's result supplies the additional constraint on their distribution relative to the measured systems. An observer's position and history can affect what is measured without imposing every possible statistical dependence.

Together with the other assumptions required for the selected Bell experiment, these conditions constrain the correlations. Violations require revising at least one premise. Determinism by itself does not imply measurement dependence, and violation of measurement independence does not by itself specify a superdeterministic mechanism. The temporal and causal account still matters.

Hall [2] supplies explicit local deterministic models using measurement dependence, including a singlet correlation model. This is established prior work, not a consequence newly derived from a simplicity prior. It shows why the task is to specify and assess the dependence rather than reject or accept it by name.

The Bell bound and the role of independence

For stochastic local responses, the conventional factorized form is

$$P(A, B \mid a, b, \lambda) = P(A \mid a, \lambda)P(B \mid b, \lambda).$$

Together with measurement independence, a normalized ensemble and the appropriate experimental setting, this restricts attainable correlations. The deterministic special case makes the CHSH bound especially transparent. For settings a,b in {0,1} and outcomes in {−1,+1}, write A_0(lambda), A_1(lambda), B_0(lambda), B_1(lambda). For each lambda,

$$C(\lambda) = A_0(B_0 + B_1) + A_1(B_0 - B_1).$$

Exactly one parenthesis vanishes and the other is either +2 or −2. Thus |C(lambda)| = 2. Averaging all four correlations with the same normalized rho(lambda) gives

$$|E_{00} + E_{01} + E_{10} - E_{11}| \leq 2.$$

The use of the same distribution for all four setting pairs is the point at which measurement independence enters this derivation. With rho(lambda|a,b), the four correlations need not be averages over the same ensemble, so this step no longer establishes the bound.

Now retain the author's preferred requirements: definite outcomes, local response functions, empirical agreement with Bell violations, and the stated sampling framework. Independence cannot remain alongside them. That is the affirmative argument for investigating measurement dependence. It is stronger and more precise than claiming that nothing in one universe can ever be independent of anything else.

3. Which variables must depend on which?

Let U be uniform on {0,1,2,3}. Define a = U mod 2 and lambda = floor(U/2). Each pair (a,lambda) occurs once, so P(a,lambda) = 1/4 = P(a)P(lambda). Both variables are deterministic functions of the same U, yet they are independent under the stated ensemble.

Set lambda = a instead, with a balanced. Then P(a = 0,lambda = 0) = 1/2, not 1/4. Both constructions have short descriptions. Common origin and shortness select neither independence nor dependence.

Apparent randomness does not decide either relationship. Digits of pi are fixed by a short algorithm, although their full statistical properties, including normality, are not proved. A pseudorandom setting generator is also a deterministic

physical system. Its outputs can look random while being independent of one variable and dependent on another. Alice's setting a and Bob's setting b can be mutually independent even when the pair is correlated with lambda. Bell's measurement independence concerns P(lambda | a,b), not simply P(a,b). A local deterministic model that also retains measurement independence still obeys the relevant Bell bound.

A single fixed history is not itself an ensemble distribution. The distribution might describe repeated trials, uncertainty about initial data or another explicitly defined measure. A claim that uniform initial conditions generically produce independence must specify that space and measure. Uniformity on a finite product space gives independent coordinate projections; a different mapping or conditioning can produce dependence. There is no universal uniform measure on every possible physical initial condition.

The block description gives the common generator proposal its ontological setting. The experimenter and measured system have histories within the same structure. The Bell calculation below distinguishes that shared membership from the statistical law needed to recover the correlations. The affirmative argument combines this internal account of experiments with the locality and outcome commitments in Section 2.

Superdeterminism is not simply another word for universal determination. It concerns dependence between settings and underlying states in a specified statistical account. Universal unitary evolution and the Standard Model do not supply that joint distribution merely by being mathematical. Everettian and superdeterministic approaches therefore are not two conclusions automatically delivered by one equation. They make different commitments about outcomes and their explanation.

Removing fundamental randomness yields determinism only when the specified laws and state fix the evolution. Removing a physical collapse law also does not entail measurement dependence. Standard Bohmian mechanics supplies a counterexample: deterministic configurations follow a nonlocal guidance law, with no fundamental collapse, while settings independence can hold for the relevant system variables [4, 6]. Everett supplies a different account without collapse. These alternatives establish that the proposed implication needs another premise; they need not be the preferred theories to serve as counterexamples.

The sharper implication follows within the Bell framework. For a hidden variable account with definite outcomes, retain local factorization and the other conditions under which a valid Bell inequality applies. If that account reproduces a violation, measurement independence must fail. This is a logical consequence of the theorem. Calling the dependence superdeterministic further specifies how it arises, commonly through prior causes or constraints; the inequality alone does not choose that causal explanation [7]. The relevant hidden variables describe the measured system in the model. Including the setting registers inside a definition of the entire universe's state would make dependence trivial without

explaining the correlations.

Bohmian mechanics supplies a useful comparator. Its configuration and nonlocal guidance law offer definite outcomes through a different causal account. The preference argued here is to investigate whether local common generation achieves the same predictive work more economically. A shorter complete description with equal likelihood has the prior advantage specified in Section 5. This comparison gives each added mechanism an explanatory task.

Einstein, Bell and the block account

EPR supplies the historical motivation: a demand for completeness tied to reasoning about separated systems. Einstein died before Bell's 1964 result. It is reasonable to ask how his commitments bear on the later choices. It is not reasonable to infer exactly what he would have accepted, or what he never considered, without the archival evidence required for those claims.

The retrospective connection is therefore ours. A block perspective includes the setting apparatus within the same world as the particles. Bell identifies conditions a local completion must satisfy. Putting these together motivates an internal physical account of experimental settings; it does not establish that Einstein unknowingly proved measurement dependence.

Bell's theorem likewise should not be explained away as a failure to understand relativity or an emotional discomfort mistaken for proof. Its mathematical content survives the independence challenge: relaxing an assumption is one of the possibilities the theorem makes explicit. A criticism of blanket dismissals of superdeterminism can stand without attributing Newtonian misunderstandings to Bell or a single psychology to the physics community.

4. A deterministic construction and its correlation measure

A finite construction makes the common generator proposal concrete. It produces setting dependence with no fundamental stochastic selection. Its purpose is to isolate the separate roles of the response rule and the distribution of preparations, using an elementary illustration rather than a proposed complete quantum theory.

Let the deterministic trial counter n cycle through 0 to 7. Set a = floor(n/4), b = floor(n/2) mod 2 and r = n mod 2. Over one complete cycle all eight triples occur once, so the empirical setting distribution has independent balanced a and b. The source supplies these settings to the devices and gives the measured systems lambda = (a,b,r). Define A(a,lambda) = r and B(b,lambda) = r XOR (lambda_a AND b). Each response uses its local setting and lambda. On the supported trials A XOR B = ab. The settings are mutually independent, but lambda is correlated with them. All of this is generated by a counter.

Write E_ab for the mean of $(-1)^{\wedge}$(A XOR B), conditional on settings a,b. Then E_00 = E_01 = E_10 = 1 and E_11 = −1. The CHSH expression E_00 +

E_01 + E_10 − E_11 is 4. Each local outcome is still balanced because r is fair. The observable marginals do not depend on the remote setting, even though measurement independence fails.

The result exceeds the quantum CHSH maximum 2 sqrt(2). Thus a very short, nonsignalling, measurement dependent rule can be empirically wrong for quantum experiments. Compactness and nonsignalling together are insufficient to recover quantum theory.

Extend lambda with a bit e, independent of a,b,r in the declared ensemble, with P(e = 1) = q. Replace B by B XOR e. The responses remain deterministic given lambda. Now E_ab = (1 − 2q)(−1)^ab and CHSH = 4(1 − 2q). Choosing q = (2 − sqrt(2))/4 reproduces 2 sqrt(2) and the four correlators of a suitable maximally violating quantum configuration. This is a choice of ensemble weight, not a derivation of it from the response functions.

The ensemble weight q need not represent fundamental chance. For a computable q in [0,1], assign successive eight trial blocks the bit e_k = floor((k+1)q) − floor(kq), starting at k = 0. Each block contains every triple (a,b,r), while e_k is fixed throughout that block. After N blocks the fraction with e = 1 is floor(Nq)/N, by telescoping. The CHSH value is therefore 4(1 − 2 floor(Nq)/N), converging to 4(1 − 2q). Thus a wholly deterministic schedule supplies the stated limiting correlations. The schedule explains how the frequencies can be generated; choosing the quantum value of q still requires a physical explanation.

Even this match addresses only the specified four setting pairs. It does not recover the full angle dependence, arbitrary preparations, sequential measurements or field theory. The example locates the work a complete generator must do: it must explain the appropriate measure and generalization, not merely display a short local response rule.

5. Compare predictive models with predictive models

Let M include the state space, dynamics, preparation distribution, observation map and experimental protocol needed to assign probabilities to a dataset D. Fix an effective encoding convention U. A possible prior is w(M) proportional to 2^(−K_U(M)), with normalization and model class stated where needed. This is a proposed model prior. Its identification with a physical measure over realized worlds is an additional premise.

The posterior comparison is

P(M_1 | D) / P(M_2 | D) = [w(M_1) / w(M_2)] [P(D | M_1) / P(D | M_2)].

A delta bit advantage gives the factor 2^delta in the chosen description weights. Ten bits gives 1,024. A likelihood disadvantage of 2,048 gives posterior odds 1:2 despite that advantage. When the models give identical likelihoods on the compared evidence, the prior advantage remains. It is not cancelled merely

because the factor is large; it is conditional on the actual encoding and model comparison.

Kolmogorov complexity is not generally computable. A supplied implementation gives an upper bound, not proof of the shortest possible code. Changing the reference machine changes complexities by bounded additive constants, but those constants can change finite model rankings. Runtime and resources are further criteria if the task is prediction with feasible computation.

Nor is K(L,I,O) generally equal to K(L) + K(I) + K(O). Shared information can be encoded jointly. A conditional description avoids paying again for output already generated by the laws and input. Conversely, omitting input precision, a state selector or a preparation measure can make an incomplete account look artificially short.

Solomonoff's universal constructions have precise dominance properties over the relevant effective classes [3]. That does not establish a unique reference machine, a physical prior over all structures or the posterior superiority of a particular quantum interpretation.

6. A stochastic law is not a list of its outcomes

The position defended here demands an account of why a particular event occurs. A probability distribution answers a different question: how outcomes are weighted under specified conditions. If its selection is declared irreducible, a deterministic explanation of that selection has been set aside. I favor searching for the underlying generator before accepting that stopping point. This is a demand for causal completeness, not a contradiction in the probability axioms. If every event must have sufficient conditions fixing it, fundamental chance is excluded by that premise; the premise itself is the philosophical issue to defend.

Measurement independence likewise is a substantive modelling assumption. Deterministic setting devices do not logically require it, and apparent randomness does not establish it. The programme therefore permits a joint account of settings and hidden data from the outset. The Bell result explains when dependence becomes necessary for a local account with definite outcomes. This gives the causal completeness commitment a definite physical target.

A fair coin law has a short specification. For N tosses it assigns each specified string probability $2^{\wedge}(-N)$. Describing one particular string exactly may require about N bits, but that is a data cost. It is not an additional N bit law hidden inside the coin model.

An all zero string also has probability $2^{\wedge}(-N)$ under that law and a short exact description given N. Therefore even the assertion that every randomly produced finite record is incompressible is false. Typicality, probability and individual description length are related but distinct notions.

A finite binary measurement outcome never needs infinitely many bits merely

because it was stochastic. Ideal continuous models raise questions about infinite precision, but laboratory observations have finite precision, and deterministic continuum models can also place unlimited information in their initial data. The comparison must treat both sides at the same observational resolution.

An Everettian model must account for the weights of the observations an observer can expect. A deterministic model with unknown initial conditions also needs a distribution over the possibilities used in prediction. For any accounts assigning the same probabilities to D, their likelihood terms agree. Giving one account a data cost while giving its rival none would compare different explanatory tasks.

One may instead impose a requirement that each realized history have a finite deterministic generator. That excludes some stochastic realizations by the chosen requirement, not by a contradiction within probability theory. The requirement is a substantive constructive premise and must be defended separately.

7. Existing proposals and their limits

Ciepielewski, Okon and Sudarsky [4] construct a local pilot wave account with specially homogeneous initial data and discuss constrained and relaxing variants. This establishes that a compact constraint can express a highly coordinated preparation. The proposed relaxation dynamics leaves competition between terms unresolved, so generic convergence cannot yet be used as the explanation of that preparation. The useful next step is to distinguish what the compact constraint already supplies from what the dynamics must establish.

Palmer [5] argues that the geometry and metric of admissible states alter the assessment of counterfactual perturbations. The proposal deserves evaluation on those terms. A metric defines distances; a probability measure assigns weights. Changing the former does not by itself derive the latter, establish typicality or prove that a complete model has low description cost.

An attractor, where established, also does not erase every initial condition cost. A basin, a convergence rate, tolerance and observation time must be specified. Convergence of a coarse observable need not select a unique microscopic history. These distinctions apply to any proposed relaxation mechanism, including proposals outside superdeterminism.

8. The stronger fine tuning objection

Wood and Spekkens [1] connect Bell correlations with faithfulness in classical causal models: observed independence can require restrictions on causal parameters beyond those encoded by the graph. A short formula may impose such a restriction and still be fine tuned in that technical sense. Replacing their criterion with low K does not refute their result.

A useful answer must explain why the permitted physical variations preserve

the observed independence and quantum statistics. One option is an independently motivated law restricting the parameter family. Another is a dynamical mechanism with a specified preparation measure. Each option has to carry the observational work it claims.

The finite example in Section 4 makes the distinction concrete. A balanced common bit makes both local outcomes unbiased. If its distribution becomes biased, Bob's marginal at $b = 1$ generally depends on a, because B then switches between r and its complement. Whether that perturbation is admissible belongs to the model. Calling the fair bit simple does not answer the perturbation question.

There is consequently no useful universal definition equating conspiracy with a large K(correlation | dynamics). Low conditional K can accompany a fragile cancellation. A complex but independently justified preparation need not be conspiratorial. Coding cost is one diagnostic alongside causal explanation, robustness and empirical fit.

9. Scientific inference and experimental variations

Science does not require settings to be chosen by an uncaused will. It does require a justified route from actual samples and interventions to the conclusions drawn from them. A measurement dependent model must explain that route and identify which experimental variations preserve its predictions.

Specify the setting generator, source preparation and data selection procedure. Predict what changes when the setting protocol is altered. Evaluate joint distributions where appropriate: conditioning on settings does not authorize omitting the very dependencies the model proposes. A common generator selected after seeing each dataset can fit observations without explaining why it should predict new ones.

The empirical target includes repeated preparations, nonsignalling marginals, the angle dependence of correlations and consistency with other physical observations. Recovery of a single Bell value is only an initial demonstration. A universal rejection of superdeterminism is not established here; neither is a blanket exemption from ordinary predictive scrutiny.

10. Conclusion

Superdeterminism seeks a common physical explanation for the source, the setting devices and their correlations. A short rule can produce a highly specific history without listing its events one by one. Apparent randomness therefore does not establish the absence of an underlying generator.

The worked example separates two kinds of independence. Settings can be independent of each other while jointly depending on hidden source information. For the stated local model with definite outcomes, the latter dependence is needed

to reproduce Bell violations. A complete theory must explain its preparation rule and recover quantum probabilities across experiments.

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